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The Paleocene Shammar Reservoir: Explore a Sub-Seismic Stratigraphic Trap to Turn an Invisible Thin Sandstone into a Robust Low UTC Opportunity in North Oman

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Abstract

Approximately a decade ago, Shammar play was discovered in the northwest of the PDO concession Block-6 which was believed to be a strong top seal for the locally sub-cropping Cretaceous Shuaiba Formation reservoirs. The commercial rates of oil production from the Shammar developed fields and its low UTC characteristic made the play very attractive to expand its regional understanding and extend the exploration activities to other areas. Therefore, a comprehensive regional evaluation conducted around the recent Shammar discoveries and in underexplored segments, delivered a robust regional geological model, which led to a new drilling campaign resulted in discovery of two fields in North Oman in 2024. Despite the resounding success, there are plenty of challenges associated with exploring the stratigraphically trapped thin sandstones of base Shammar including: sub-seismic nature and lack of acoustic impedance control, prediction of reservoir occurrence, thickness, lateral distribution and extension and well geo-steering.

In addition to the multi-disciplinary integration of all data including sedimentology (cuttings, SWS and conventional core), stratigraphy, petrophysics, seismic data, wells and geological models, the key enabler for the exploration success in the recent Shammar discoveries was the introduction of horizontal drilling at Shammar level. Two wells were geosteered horizontally successfully, marked as the first exploration Shammar horizontal wells and led to two new discoveries. Shammar accommodation and reservoir thicknesses heterogeneities are well known to be unpredictable. Horizontal drilling was utilized to de-risk additional subsurface points 800-1000m away from the main hole, which reduced the uncertainties and helped to understand the extension of the sand fairway.

As the company continues to focus on resilient and attractive projects, the Shammar as shallow clastic play will continue to play an important role in sustaining exploration and development activities. Building on the success of this stratigraphic play concept, a Shammar play regional evaluation is ongoing to identify new concepts and leads. These volumes will support the exploration drilling plans in the upcoming 2-3 years. The concept is also being investigated over other areas where seismic acquisition or processing is underway. These will further contribute to the future business opportunities for the company and the Sultanate of Oman. The remaining Shammar prospectivity in North Oman represents a high-risk but potentially high-reward

exploration target, given its location in an underexplored area. With the need for multiple wells, operational efficiency is a key to minimize financial exposure, especially when considering the unpredictability of the play.

Introduction

The Shammar represents the lowest member of the Umm er Radhuma (UER) Formation, delineated in Oman's subsurface as the UER Shammar (JRRHS) Shale unit. It exhibits an eastward pinch-out, with maximum recorded thickness observed in the northern sector and progressive thinning towards central and southern Oman. Stratigraphically, the Shammar forms the oldest of three members of the Umm er Radhuma Formation, which unconformably overlies the Shuaiba, Nahr Umr, or Natih subcropping formations on a regional scale (Figure 1). In the Greater Lekhwair area of northern Oman, the basal Shammar reservoir is characterized as a heterolithic clastic facies interbedded with sandstones and/or conglomerates. Extensive core acquisition and detailed petrophysical analysis conducted by PDO over recent years have provided comprehensive reservoir characterization. Well log data and core measurements reveal significant lateral thickness variability of the basal Shammar reservoir within Greater Lekhwair, ranging from decimeters up to 10 meters. The Shammar reservoir section can be subdivided in 3 sub-units from base to top including:

- UNIT 1: Conglomerates, fining into sandstones and carbonaceous shales, low facies variability, confined, variable reservoir properties, pay zone with the best reservoir quality.
- UNIT 2 Sandstone, fining occasionally into bioturbated heterolithics and claystones, rarely conglomerate, high facies variability, semi-confined, pay zone with variable mobility & reservoir quality
- UNIT 3: Argillaceous and calcareous bioturbated bioclastic sandstones (to sandy limestones in rare cases), moderate facies variability, unconfined, variable reservoir properties, oil stained but rarely pay zone.

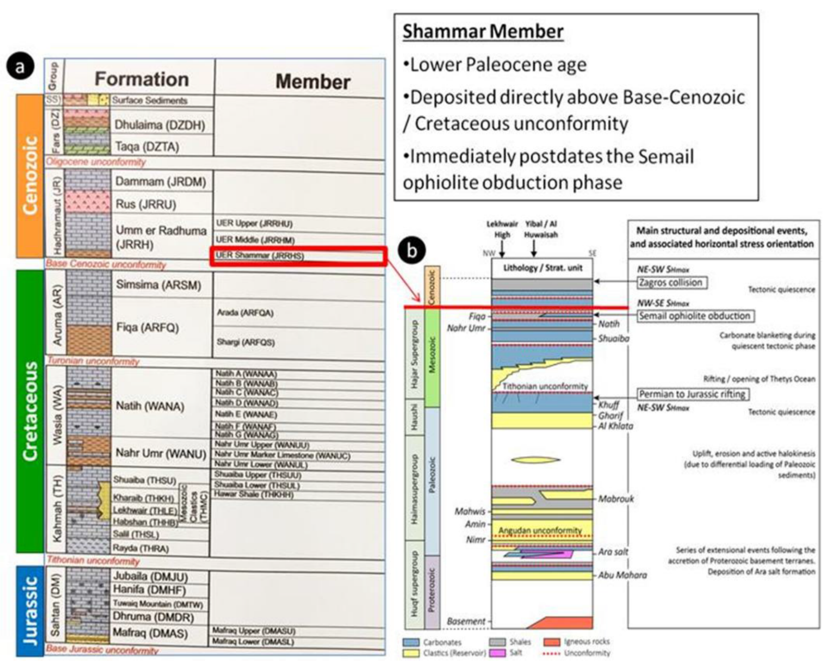

Figure 1—(a) Stratigraphic column (after Forbes et al., 2010) showing the position of the Shammar Member as part of the Paleocene Umm er Radhuma Formation. (b) Tectonostratigraphic sketch showing the emplacement of the Shammar Member, postdating immediately the Semail Ophiolite obduction episode (from Bazalgette and Salem, 2018, modified from al Kindi and Richard, 2014).

The most recent regional stratigraphic and ichnological analysis of the Shammar Formation corroborates its deposition within an incised valley system developed under a transgressive systems tract. Accommodation space for the Lower Shammar is attributed to erosional truncation of the mechanically weak Nahr Umr subcrop, preferentially occurring within structural depressions. The alignment of the incised valley within these lows outlines the prospective play fairways. The lower Shammar claystone unit is a thick succession of dark, laminated pyritic claystones overlies the reservoir, followed by a correlatable carbonate marker bed (upper Marker bed) that is extensive laterally above the shales, ~3-5m from the top of the reservoir. It shows a nodular fabric due to bioturbation, and fossil content (foraminifera, bioclasts) indicating deposition in open, normal marine, low energy environments, of a mid to distal carbonate ramp. This carbonate unit separates the lower Shammar claystone from the Upper Shammar claystone (Figure 2):

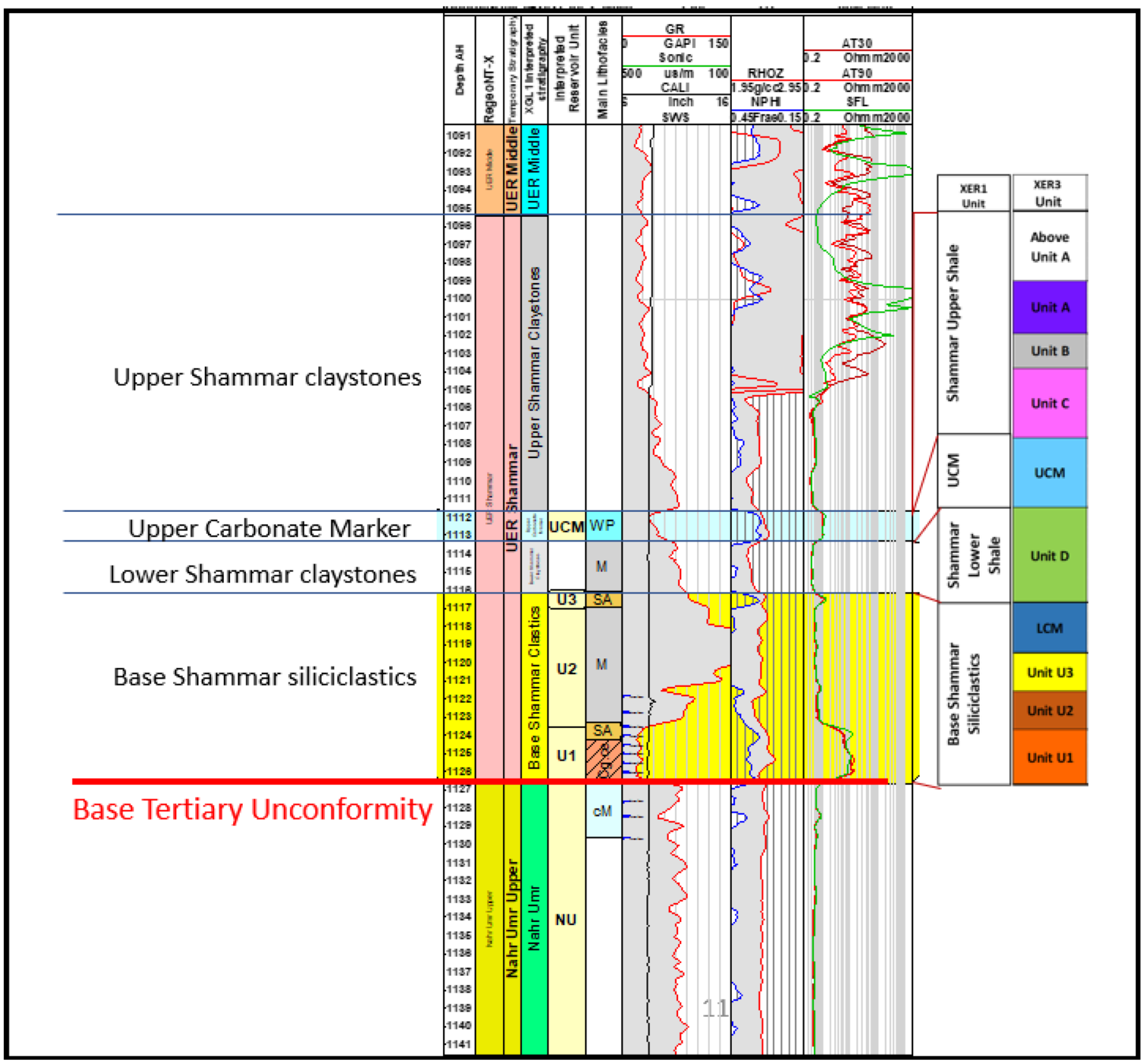

Figure 2—Stratigraphic column showing the main subdivisions of the Shammar Member.

Exploration History Background

Hydrocarbon presence within the Shammar reservoir of PDO Block-6 was initially identified in 2014, following a pressure influx during drilling operations through the "Shammar shale" in an exploration well targeting deeper objectives. The influx led to oil observations at the surface, prompting a focused exploration and appraisal drilling campaign the following year. This effort, targeting the relatively thin Shammar interval, culminated in a successful discovery. Building on the encouraging outcomes—including high initial oil production rates, attributed to favorable reservoir quality and excellent fluid characteristics with no associated water production—a Phase 2 exploration program was launched. This subsequent campaign confirmed the extension of Shammar reservoir prospectivity into the southern sector of the field, leading to additional volume bookings within the Shammar play.

Following the discovery of the two Shammar fields, the PDO exploration team undertook a comprehensive regional, multidisciplinary study focused on the vicinity of the recent Shammar discoveries and adjacent underexplored sectors (2021-2022). The objective was to assess the potential lateral extension of the Shammar reservoir across the northwestern portion of the PDO Block-6 concession. This effort aimed to identify additional incised valley and channel systems, thereby introducing new exploration concepts and generating prospective leads for the broader Shammar play. The outcomes of this study are expected to significantly optimize the planning of future exploration and development drilling activities within the North Oman Shammar oil Play. The study employed an integrated geoscientific approach, incorporating core and cuttings analysis, petrophysical log-based facies interpretation, sedimentological and stratigraphic frameworks, seismic attribute analysis, and Gross Depositional Environment (GDE) mapping. This multi-disciplinary integration enhanced the geological understanding of the Shammar system and expanded the exploration portfolio with refined subsurface models.

Methods and Early Results

The regional evaluation update delivered a robust regional geological model with ten concepts identified of these, four were high graded based on integrated risk and value assessment and subsequently matured into drillable prospects. This led to a new drilling campaign consisting of 10 wells to explore and test models resulted in a discovery of two fields made in North Oman in 2024. The study has utilized all available data from different scales:

- ***Sedimentology:***

  An intensive sedimentological study was caried during the project evaluation utilizing Exploration and Development Shammar wells & wells with Shammar penetration drilled between 2015-2020 with all available data include: cored wells, side-wall sampled (SWS) wells, thin sections for petrography, samples for XRD mineralogy and samples for Scanning Electron Microscopy (SEM) to update the play understanding on the geological setting, units subdivision and depositional model.
- ***Stratigraphy:***

  Integrated lithostratigraphic, biostratigraphic, and chronostratigraphic analyses were conducted to accurately delineate the top and base of the Shammar Member and to establish a consistent stratigraphic framework subdivided into mappable units. Chronostratigraphic control, primarily derived from fossil assemblage analysis and the establishment of biostratigraphic zones, enabled a more robust interpretation of key stratigraphic surfaces—particularly in areas where the basal Shammar siliciclastic reservoir is overlying the shaly facies of the underlying Nahr Umr Formation. The defined biozones significantly enhanced top/base picks in intervals where the basal Shammar exhibits carbonate lithologies rather than siliciclastics, as well as in regions where the Shammar biozones remain undifferentiated and directly overlie the Shuaiba Formation. This multi-disciplinary stratigraphic approach has improved correlation and reservoir predictability.
- ***Seismic:***

  The newly acquired Lekhwair WAZ seismic dataset was utilized to refine the interpretation of the top of the Shammar Member and the base Paleocene unconformity, along with comprehensive attribute analysis. Shammar subcrop geometries were delineated by integrating Thickness Decomposition seismic attributes with existing well data. Utilizing established accommodation space controls—derived from well parameters—potential sand fairways were mapped through the application of Wide Band Spectral Decomposition (WBSD) seismic attributes. Furthermore, new incised valley and channel systems were identified on the WBSD data from the Lekhwair WAZ survey, with subsequent validation provided by well correlations. QI trials made to sweetspot the Shammar reservoir were not successful. The outcome was inconsistent with well observations and results.

- ***Well Data***

  Mapping of the basal Shammar clastic interval within the Greater Lekhwair area indicates that reservoir distribution is predominantly restricted to regions where the Nahr Umr Formation subcrops, as supported by well control. The Nahr Umr shales, characterized by lower mechanical competence, are interpreted to be more susceptible to erosional incision, thereby facilitating the development of incised valleys and associated accommodation space for the deposition of Lower Shammar clastics. The planform geometry and sinuosity of these incised valleys appear to be influenced by the underlying, more erosion-resistant Shuaiba and potentially Natih carbonate platforms at the Base Cenozoic subcrop, which may act as structural or stratigraphic barriers, locally inhibiting valley incision. Additionally, well data suggest that areas underlain by a relatively thicker Nahr Umr subcrop correspond to zones with absent or thin Shammar clastic deposits, implying that variations in Nahr Umr thickness may exert a local control on sand fairway development and reservoir continuity.

Early Results from Regional Evaluation

- Possible new incised valleys extending and branching were mapped in the AOI.
- Locally thicker Nahr Umr subcrop might have some control to the sand fairways distribution.
- Ten concepts were identified that later shaped the recent successful drilling campaign.
- Expected reservoir units in these concepts are generally unit 1 and 2.
- Unit 3 is expected to be more laterally extensive possibly improving in quality towards the Saudi-Oman border. Lateral extension of Units 1 & 2 is uncertain away from well control.
- Main Risk: reservoir presence and lateral and bottom seal where area's with Natih is the sub-crop.

Exploration drilling strategy

Building on the outcomes of the regional evaluation and the development of a comprehensive depositional and geological model, several new stratigraphic play concepts and leads were identified across the Shammar play. These insights provided the foundation for a proposed exploration campaign, which received endorsement from both the Technical Assurance Team and senior management. The campaign comprises multiple exploration wells designed to test the validity of the geological model, de-risk the identified leads, reduce subsurface uncertainty, and ultimately demonstrate commercial viability. To enhance the probability of technical and commercial success, total of 10 wells distributed into 4 segments (leads). The intense appraisal campaign caters for the complexity of the play in terms of unpredictable distribution and reservoir heterogeneity. The wells were placed based on seismic features on WBSD, down dip from the Shuaiba highs, drill above Shuaiba grabens where Shammar accommodation (and hence sand presence) is inferred, away from thicker Nahr Ummar "uneroded" and on local gentle sags or small undulations of the base Shammar in seismic section. In addition, a contingent plan for sidetrack based on the thickness of base Shammar accommodation space and deepening some wells to un-lock potential deeper reservoir endorsed, to maximize the value of the campaign and add a control point for Exploration and development teams.

The exploration campaign proved highly successful, resulting in two significant discoveries within the Shammar reservoir, both demonstrating substantial recoverable volumes and strong economic viability for development. Additionally, two deeper reservoir intervals were identified and successfully unlocked, further expanding the subsurface potential of the area. A phased development strategy has been initiated, incorporating early monetization of the exploration and appraisal wells, thereby aligning with PDO's broader production growth objectives. The outcomes of the campaign have effectively validated both the stratigraphic play concept and the commercial viability of the Shammar reservoir within Areas A and B, marking a key milestone in the maturation of this emerging play (Figure 3).

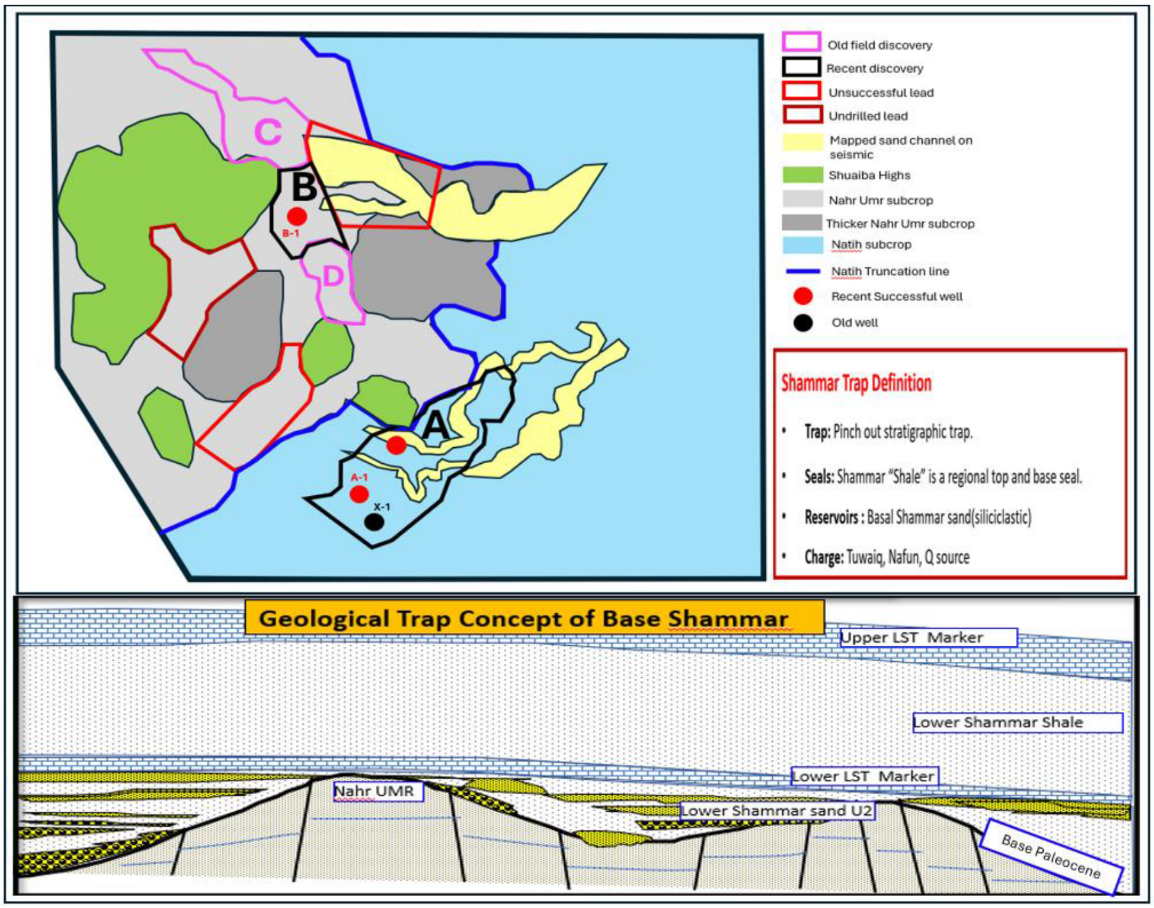

Figure 3—The geological model of base Shammar reservoir in the area of interest.

Case Study – Discovery A

The A prospect was prioritized, where Unit-3 was identified as a higher-quality, productive reservoir. Updated GDE and mapping of Unit-3 indicated the presence of laterally extensive, high-quality sands—potentially indicative of a shoreface depositional setting. Supporting evidence also came from legacy well X-1, which, though drilled for deeper targets, exhibited high resistivity over a potential Shammar reservoir interval, implying the presence of hydrocarbon-bearing sand. The primary geological risk identified for the A opportunity was the integrity of the base seal, given that Shammar clastics directly overlie the carbonate subcrop of the Natih Formation, raising concerns about bottom seal effectiveness.

The well A-1 was subsequently drilled in late 2023, encountering 1.4 meters of net oil pay at the base of the Shammar reservoir, with an average porosity of 26% and hydrocarbon saturation of 61%. To further evaluate lateral continuity and reservoir quality, the well was sidetracked and successfully penetrated over 140 meters of net sand. Well testing confirmed a stable commercial production rate, with no water production. These results substantiate the presence of a clastic reservoir within this segment, in alignment with pre-drill predictions derived from the regional geological model, thereby de-risking the play and validating the depositional and stratigraphic interpretation in the area.

Case Study – Discovery B

The B opportunity was principally derived from its favorable structural position between two established Shammar discoveries—Fields C and D—and supported by the GDE map, which suggested the potential presence of a sand fairway connecting the two accumulations. This interpretation positioned the B area as a possible stratigraphic and depositional linkage zone, warranting further exploration.

Well B-1 was drilled to test the southern extension of the sand fairway from Field C and to delineate its continuity. The pilot hole encountered 4.6 meters of accommodation space but did not penetrate any reservoir-quality sand. The well was subsequently sidetracked in a horizontal trajectory, where it intersected approximately 230 meters of good-quality Shammar sands, distributed across two discrete sand bodies. Reservoir evaluation indicated average porosity of ~20% and oil saturation of ~65%. Well testing demonstrated a high commercial oil production rate, with no water production—attributed to favorable reservoir properties and excellent fluid quality. These results confirm the extension of the sand fairway into the B area, supporting the pre-drill depositional model and enhancing confidence in the stratigraphic continuity across the play.

Shammar Project Challenges

Despite the resounding success, there are several challenges associated with exploring the stratigraphically trapped thin sandstones of base Shammar including:

- ***Sub-seismic nature:***
  Seismic resolution remains a limiting factor in delineating the Shammar reservoir, as the reservoir interval is frequently thinner than the vertical resolution limit of the available seismic data. Vertical Seismic Profile (VSP) analysis indicates that even high-frequency reflectivity data (~80 Hz) are insufficient to resolve the thin Shammar sands and associated accommodation space, particularly in areas where net sand thickness is minimal. This limitation posed significant challenges during the early phases of exploration and development, where reliable seismic interpretation of the reservoir was difficult. The seismic character of the Top Shammar and the Base Paleocene Unconformity exhibits notable lateral variability across wells, further complicating consistent horizon picking and stratigraphic correlation. As a result, early mapping efforts relied heavily on well calibration, attribute analysis, and integration of multiple datasets to constrain the reservoir geometry and extent.
- ***Prediction of reservoir occurrence, thickness, lateral distribution and extension:***
  The predictive accuracy of sand distribution within the Shammar reservoir remains limited, resulting in a high risk of encountering dry wells. This is primarily due to persistent uncertainties in the depositional model and the lateral and vertical connectivity of individual sand units. The complexity is further compounded by the fine-scale heterogeneity of the reservoir, where lithofacies units can be only a few centimeters thick—below the vertical resolution of conventional log responses—making accurate lithological interpretation and facies differentiation highly challenging. On both regional and field scales, the Shammar reservoir continues to demonstrate high geological uncertainty. However, the evolving understanding of its depositional environment provides a valuable basis for improved predictive modeling of sand fairways between well penetrations. Integration of sedimentological interpretation, facies modeling, and well-seismic correlation will be essential to reduce risk and enhance reservoir development strategies.

- ***Well geo-steering:***

  The nature of base Shammar clastic reservoir (namely its thin, laterally heterogeneous, and geologically unpredictable distribution) pose significant challenges for optimal well placement and effective geosteering. The maximum horizontal section drilled to date within the Shammar reservoir has reached 1,200 meters, while the longest continuous sandstone interval encountered measures approximately 600 meters. These drilling metrics highlight the complexity of maintaining wellbore trajectory within the targeted reservoir zone. Through close collaboration and integration between PDO's exploration and development teams, a standardized geosteering workflow has been established and routinely applied to horizontal well delivery within the Shammar play (Figure 4). This workflow enables safe well execution while optimizing reservoir exposure and meeting key subsurface objectives. The Geosteering decisions are made based on real time logging data and cuttings (one practice is by differentiating Shammar grey shale from olive green Nahr Umr shale). For landing the well it is recommended while landing to follow the updated well plan (from H1) and hold at 83 degrees till confirm the Upper Marker Limestone (UMLS). Top Lower Marker Limestone (LMLS) should be revised after hitting UML. Target to hit LMLS with inclination of ~ 87 degrees. Then build to 89.5-90 deg to land the well and monitoring typical responses for both good quality Shammer sandstone and Nahr Umr. The challenges associated with well placement and geo-steering in base Shammar sand is mainly due to the complex depositional reservoir architecture, accommodation space thickness and sand presence/continuity, the unconformity nature of Base Paleocene, the high risk of drilling in the reactive Nahr Umr shale and the Dogleg severity limitation. The impact of these can be minimized by maturing the reservoir geology knowledge/ understanding and the integration between geologists and well placement engineers to overcome the challenges while drilling and developing the very thin sandstone reservoir.

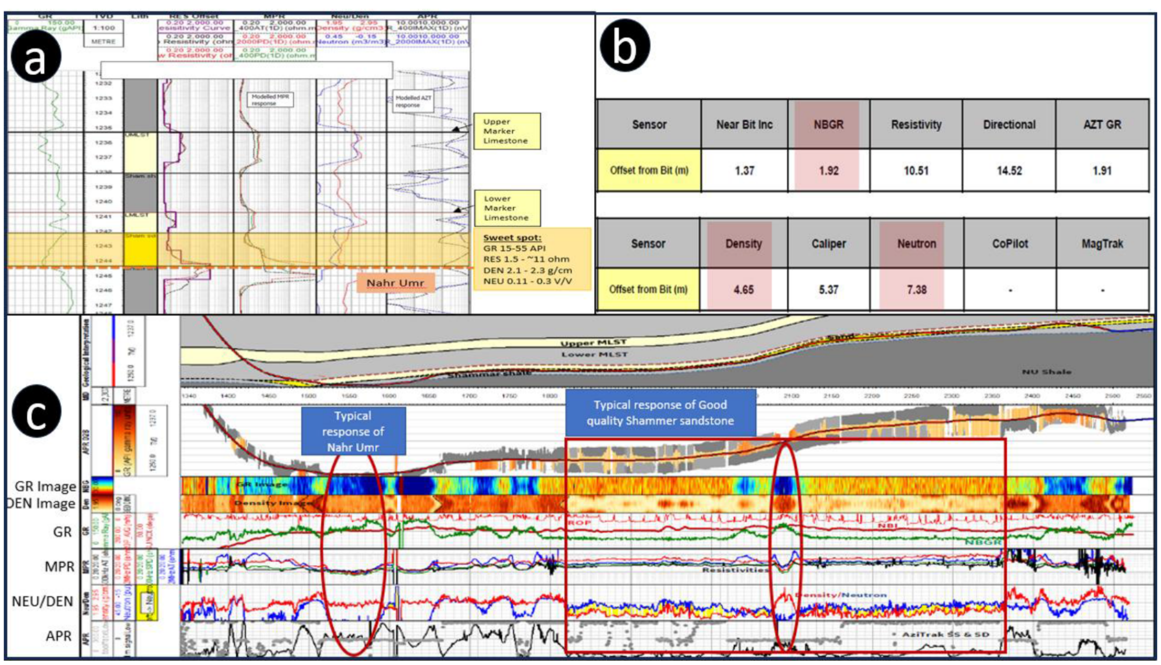

Figure 4—(a) Key markers and their typical log response that usually use for well placement of Shammar wells. (b) Typical set-up of sensors offset from Bit for Shammar well geo-steering. (c) Typical logs response of good quality Shammer sandstone and Nahr Umr Shale.

Way Forward and Follow up activity

As the company maintains its focus on resilient, high-value projects, the Shammar shallow clastic play continues to represent a strategically important component of ongoing exploration and development efforts. Leveraging the demonstrated success of the stratigraphic play concept, a new phase of regional evaluation is currently underway to identify additional leads and play concepts within the Shammar interval. These emerging volumes are expected to underpin the exploration drilling portfolio over the next 2–3 years.

Concurrently, the Shammar play is being assessed in other areas where new seismic acquisition and reprocessing initiatives are ongoing, aiming to enhance imaging quality and support further derisking of the play. These efforts will contribute to the generation of future business opportunities for the company and align with the broader hydrocarbon development strategy of the Sultanate of Oman.

Despite its promise, the remaining Shammar prospectivity in North Oman is characterized by high geological uncertainty and exploration risk, primarily due to the complex depositional architecture. However, the potential for big commercial volumes renders it a high-reward target. Given the necessity for multiple wells to delineate and appraise this stratigraphic play, maintaining operational efficiency and cost discipline will be critical to minimizing financial exposure and ensuring project viability in this underexplored frontier.

Conclusions

Beyond the comprehensive multidisciplinary integration of sedimentology (cuttings, sidewall samples, and conventional core), stratigraphy, petrophysics, seismic interpretation, well data, and geological modeling, a critical enabler for the recent exploration success in the Shammar play was the implementation of horizontal drilling at the Shammar reservoir level.

Two horizontally geosteered exploration wells—marking the first of their kind in the Shammar stratigraphic play—were successfully drilled and directly resulted in two new field discoveries. Given the well-documented heterogeneity and lateral unpredictability of Shammar accommodation space and reservoir thickness, horizontal drilling proved instrumental in mitigating subsurface uncertainty. By extending the well trajectory 800–1,000 meters laterally from the pilot hole, these wells effectively tested long reservoir intervals and improved understanding of sand fairway continuity. This approach not only reduced geological risk but also provided critical insights into reservoir distribution and geometry, further supporting the evolving stratigraphic model of the play.

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